MILESTONE 4 · TRAINING, TUNING & EVALUATION

AI-Powered Driver Wellness & Safety Monitoring System

Five deep-learning modules — implemented, hyperparameter-tuned, trained and evaluated end-to-end, with deployable checkpoints ready for the Risk Fusion Engine.

Kushagra · Shiwani · Shubham · Sohini · Ravina

BSDA4001 · DS/AI Lab Course

MILESTONE 4 · FOCUS

From Architecture to Trained Models

Where earlier milestones prepared data and selected architectures, Milestone 4 delivers complete implementation, optimization and evaluation for every module.

Train each module

Train the selected architecture for all five modules on task-specific data.

Optimize hyperparameters

Run systematic sweeps to find the best-performing configuration per module.

Evaluate rigorously

Score with accuracy, F1, mAP, confusion matrices and per-class analysis.

Study generalization

Analyze regularization and optimization effects on validation behaviour.

Ship checkpoints

Generate deployable inference checkpoints and pipelines for every module.

Prepare integration

Ready all trained models for the unified Driver Wellness framework.

OVERALL SYSTEM

Five Models, One Wellness Score

Driver Camera → Video Stream

- **Video Fatigue**
  EfficientNet-B0 + BiLSTM
- **Landmark Fatigue**
  LSTM
- **Driver Activity**
  MobileNetV3
- **Seat Belt and Phone Usage**
  YOLOv8n
- **Smoking / Drinking**
  YOLOv8n

Driver Risk Fusion Engine · weighted scoring

Driver Wellness Assessment (0–100)

Alert Generation → Fleet Monitoring Dashboard

SHARED METHODOLOGY

One Workflow, One Environment

COMMON M4 EXPERIMENTAL WORKFLOW

1. Dataset
2. Data Preprocessing
3. Model Selection
4. **Hyperparameter Optimization**
5. Model Training
6. Validation
7. Performance Evaluation
8. **Best Checkpoint Selection**
9. Inference Pipeline

SOFTWARE ENVIRONMENT

Language	**Python**	DL Framework	**PyTorch**
Detection	**Ultralytics YOLOv8n**	Vision	**OpenCV**
Landmarks	**MediaPipe**	Eval	**Scikit-learn**

HARDWARE ENVIRONMENT

- **NVIDIA** Tesla T4 GPU
- **CUDA** PyTorch backend
- **Runtime** Google Colab
- **Python** 3.x

Same platform across all five modules keeps experimental conditions consistent and reproducible.

TEAM & SCOPE

Five Modules, Five Owners

MEMBER	MODULE	DATASET	MODEL	OUTPUT
K **Kushagra**	Video-Based Fatigue Detection	UTA-RLDD	**EfficientNet-B0 + BiLSTM**	Fatigue Level
S **Shiwani**	Landmark-Based Fatigue Detection	YawDD	**LSTM**	Drowsy/ Alert / Mild Fatigue
S **Shubham**	Driver Activity Classification	AUC Distracted	**MobileNetV3**	Driver Activity
S **Sohini**	Seat Belt and Phone Usage Detection	YOLO - DMS	**YOLOv8n**	Seat Belt / Phone Usage
R **Ravina**	Smoking & Drinking Detection	Merged YOLO	**YOLOv8n**	Smoking / Drinking

MODULE · VIDEO-BASED FATIGUE DETECTION

K Kushagra

EfficientNet-B0 + BiLSTM · UTA-RLDD

OBJECTIVE & DATA

Detects fatigue directly from video sequences, learning spatial and temporal cues — prolonged eye closure, yawning and head movement — that a single frame cannot capture.

UTA-RLDD DATASET SPLIT

46,186

Training sequences

7,042

Validation sequences

Short videos are zero-padded to maintain a fixed temporal sequence length.

PREPROCESSING

- Frame Extraction
- Uniform Sampling
- Sequence Padding
- Resize(224×224)
- ImageNet Norm
- Tensor

ARCHITECTURE

- Input Video
- **EfficientNet-B0 Backbone**
- Frame Embeddings
- **Bidirectional LSTM**
- FC Layer → Softmax
- 3-Class Fatigue Classification

K

Kushagra

Baseline training & evaluation

ARCHITECTURE CONFIG

Backbone	**EfficientNet-B0**
Temporal Layer	**Bidirectional LSTM**
Hidden Size	**256**
LSTM Layers	**1**
Dropout	**0.30**
Output Classes	**3**
Mixed Precision	**Enabled**
Gradient Scaling	**Enabled**

QUANTITATIVE RESULTS — VALIDATION

31.78%	29.20%	31.78%	28.58%
Val Accuracy	Precision	Recall	F1-score

Training Loss **0.0811**

Validation Loss **2.8596**

READING & NEXT STEP

The baseline learned spatial-temporal representations, but the large train/val loss gap and low validation accuracy show the model needs longer training and stronger regularization to generalize. Deliverable checkpoint: driver_wellness_epoch1.pth.

MODULE · LANDMARK-BASED FATIGUE DETECTION

S Shiwani

LSTM · YawDD · MediaPipe features

OBJECTIVE & FEATURES

Reads five interpretable geometric signals per frame instead of raw pixels — the same evidence a human uses to judge drowsiness — over consecutive frames.

- EAR — Eye Aspect Ratio
- MAR — Mouth Aspect Ratio
- Head Pitch
- Head Yaw
- Head Roll

The Talking_Yawning class was removed (too few samples) → three balanced classes.

PIPELINE

Frame Extraction

↓

MediaPipe Landmarker

↓

EAR/MAR/Pose

↓

Feature Cleaning

↓

Normalization

↓

Sliding Window (45)

CLASS DISTRIBUTION

Normal 105

Talking 100

Yawning 102

MODEL CONFIG

128 Hidden

2 Layers

45 Window

5 Features

S Shiwani

Tuning, evaluation & risk mapping

LSTM ARCHITECTURE

Feature Seq (45×5)

LSTM Layer 1

LSTM Layer 2

Final Hidden State

FC Layer ⟶ Softmax

Prediction

TUNING & EVALUATION

Swept learning rate, batch size, hidden size, LSTM layers, window size, dropout and weight decay for the best accuracy–generalization balance.

EVALUATED WITH

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Distinguishes Normal / Talking / Yawning from temporal facial features.

FATIGUE RISK MAPPING

A rule-based layer converts behaviour predictions into interpretable fatigue levels.

Alert
0% yawning

Mild Fatigue
0–15% yawning

Drowsy
> 15% yawning

MODULE · DRIVER ACTIVITY CLASSIFICATION

S Shubham

MobileNetV3 · AUC Distracted Driver

OBJECTIVE & DATASET

Classifies driver activity from single RGB frames to flag distraction early and support overall driver safety.

7,276

Total images

70/15/15

Train / validation split/Test spit

MODEL SELECTION

- **MobileNetV3** — Lightweight, fast inference
- **ResNet50** — High representation power
- **EfficientNet-B0** — Highest accuracy

PREPROCESSING

Driver Image
↓
Resize 224×224
↓
Augmentation
↓
ImageNet Norm
↓
Tensor
↓
Model Input

ARCHITECTURE

RGB Driver Image

MobileNetV3 Backbone

Global Avg Pooling

FC Layer

Softmax

Activity Prediction

S

Shubham

Hyperparameter tuning & performance

BEST CONFIGURATION

Learning rate, batch size, optimizer and epochs were swept; the winning setup:

Learning Rate	**0.0005**
Batch Size	**32**
Optimizer	**Adam**

Evaluated: accuracy, precision, recall, F1, confusion matrix.

VALIDATION ACCURACY

~94%

classification accuracy

WHY MOBILENETV3

- EfficientNet-B0 scored highest, but MobileNetV3 was chosen for the efficiency–accuracy trade-off.
- Depthwise-separable blocks keep inference fast on edge hardware (Jetson / Raspberry Pi).
- ~94% accuracy at a fraction of the compute suits real-time driver monitoring.
- Delivers a deployable checkpoint plus evaluation plots and confusion matrix.

MODULE · SEAT BELT & PHONE USAGE DETECTION

S Sohini

YOLOv8n | Custom Driver Monitoring Dataset

OBJECTIVE

Automatically detects whether the driver is wearing a seat belt or using a phone while driving, one of the strongest indicators of driver safety, and localizes it in the cabin.

TRAINING STRATEGY

- Transfer learning from pretrained YOLO weights
- AdamW optimizer for efficient model optimization
- Mixed Precision (AMP) GPU training
- Best-checkpoint selection (best.pt)

PREPROCESSING

Input Image
↓
Resize (640 x 640)
↓
Annotation Verification
↓
YOLO Label Format
↓
YOLOv8 Built-in Augmentation
↓
Model Input

DETECTOR ARCHITECTURE

Driver Image
|
YOLO Backbone
|
PAN-FPN Neck
|
Detection Head
|
Bounding Boxes
|
Seat Belt and Phone Usage Detection

S Sohini

Evaluation & deliverables

EVALUATION METRICS

Automatically detects seat belt usage and mobile phone usage inside the vehicle cabin to support real-time driver safety monitoring.

mAP@0.5 (95.26%)	**Precision (92.92%)**	**Recall (91.21%)**	**mAP@0.5:0.95 (66.62%)**

OUTCOME

The trained YOLOv8n detector achieved high precision and recall, providing reliable real-time seat belt and phone usage detection suitable for integration into the Driver Wellness monitoring.

MODULE DELIVERABLES

- **best.pt checkpoint**
- **Training notebook**
- **Validation predictions**
- **Detection visualizations**
- **Performance metrics**

MODULE · SMOKING & DRINKING DETECTION

R Ravina

YOLOv8n · guided hyperparameter tuning

APPROACH

A two-class in-cabin detector (smoking, drinking) at 640×640. M4 re-establishes a clean YOLOv8n baseline, then runs a guided six-trial grid over the highest-leverage fine-tuning knobs.

DATASET (YOLO FORMAT)

8,884	370	371
Train images	Val images	Test images

Classes are near-balanced in every split, so the smoking-vs-drinking gap is a difficulty effect, not imbalance.

COCO-pretrained yolov8n.pt · 319/355 tensors transfer · head → 2 classes.

GUIDED GRID — 6 TRIALS

- **t1** lr .01 · SGD
- **t2** lr .005 · SGD · cos
- **t3** lr .001 · AdamW · cos
- **t4** lr .005 · AdamW · wd
- **t5** lr .01 · SGD · no-mos
- **t6** lr .001 · SGD · wd

Each screened for 30 epochs on full train data; winner picked by val mAP@50-95.

WINNING CONFIG

t3_lr001_adamw

lr0	**0.001**
optimizer	**AdamW**
cos_lr	**True**
close_mosaic	**10**
imgsz	**640**
epochs	**80**

Retrained fresh from pretrained weights for 80 epochs (~3.46 h on T4).

R Ravina

Held-out test performance & findings

HELD-OUT TEST — FINAL YOLOV8N

0.863	**0.771**	**0.818**	**0.451**
Precision	Recall	mAP@50	mAP@50-95

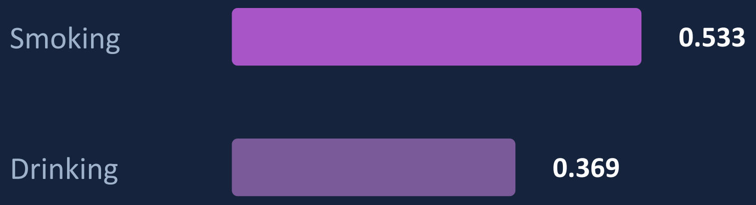
371 test images · 445 instances · slightly beats validation on the strict metric → not overfit.

TUNING GAINS OVER BASELINE (VALIDATION)

+2.50%	+2.19%	+0.48%	−2.74%
Precision	mAP@50-95	mAP@50	Recall

THE CLASS-DIFFICULTY GAP

Per-class test mAP@50-95



Drinking spans many vessels & poses — a difficulty gap, not imbalance. It is the clear target for future data work.

MODEL FOOTPRINT

3.01 M	8.1	~2.5 ms
parameters	GFLOPs	per image (T4)

COMPARATIVE ANALYSIS

Five Paradigms, One Framework

MODULE	MODEL	LEARNING TYPE	PRIMARY STRENGTH
Video Fatigue	EfficientNet-B0 + BiLSTM	Sequential	Learns long-term temporal fatigue patterns
Landmark Fatigue	LSTM	Sequential	Lightweight facial features for efficient estimation
Driver Activity	MobileNetV3	Image Classification	Accurate real-time behaviour recognition
Seat Belt and Phone Usage	YOLOv8n	Object Detection	Fast localization & compliance detection
Smoking & Drinking	YOLOv8n	Object Detection	Simultaneous detection of unsafe activities

Each model maximizes performance while staying computationally efficient for real-time deployment.